ECE250: Signals & Systems Quiz-6

Date: 15th October, 2025 **Total Marks:** 10 **Duration:** 15 Minutes

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Note:

1. Please provide proper mathematical justifications with your answers. No marks will be awarded without a valid justification.
2. Institute Plagiarism policy are strictly applicable.

[CO1 CO2 CO3] **[Q1]**: Consider a discrete-time LTI system whose frequency response is

$$H(e^{j\omega}) = \begin{cases} 1, & |\omega| \leq \frac{\pi}{2} \\ 0, & \frac{\pi}{2} < |\omega| < \pi \end{cases}$$

Given the input to this system is

$$x[n] = \sum_{k=-\infty}^{+\infty} \delta[n - 4k],$$

Determine the Fourier series coefficients of the output $y[n]$ and write its expression. **[5 Marks]**

[CO1 CO2 CO4] **[Q2]**: Calculate the Fourier transform of the following signal: **[3 Marks]**

$$e^{-2|t-1|}$$

Sketch and label the magnitude. **[2 Marks]**